

CUSTOMER MILESTONE BRIEF

MS #23 - Stable Composite Estate and Developer Golden Path

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.23.0
Date	2026-08-27
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #23 - Stable Composite Estate and Developer Golden Path converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Stabilizes the semantic inputs used to build the estate graph and gives developers one reproducible verification path.

Why this matters to a customer

Stabilizes the semantic inputs used to build the estate graph and gives developers one reproducible verification path.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a content-addressed semantic-input manifest that limits canonical modern-source identity to explicitly declared candidates and mappings while retaining fail-closed drift behavior.
- Added a deterministic read-only composite estate that overlays validated PL/I nodes and edges on the canonical graph without changing its identity or evidence contract.
- Added PL/I relationship ontology coverage and a complete Explorer perspective for ACCTPL1 to CBACT04C and ACCTPL1 to Db2 AUTHFRDS lineage.
- Added a separately content-addressed composite source pack so PL/I evidence excerpts are inspectable without modifying the canonical evidence pack.

- Preserved canonical runtime and audit bindings inside the composite Explorer and added an explicit UI banner showing canonical/composite hashes and the non-equivalence claim boundary.
- Added doctor, demo, explorer, and verify developer entry points for POSIX and PowerShell, with required-versus-optional prerequisite diagnostics and actionable remediation.
- Added fail-closed tests for semantic input drift, undeclared implementation changes, fragment drift, projection tampering, cross-estate navigation, and canonical runtime/audit compatibility.
- Kept gates 6 and 8 blocked, gate 7 mechanism-ready, mainframe_equivalent: false, and production_ready: false.

How it differs from earlier work

MS #23 builds on MS #22 (Mixed PL/I Modernization Proof). The earlier milestone established its own bounded capability; this milestone adds stable composite estate and developer golden path while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use stable composite estate and developer golden path as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.23.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.